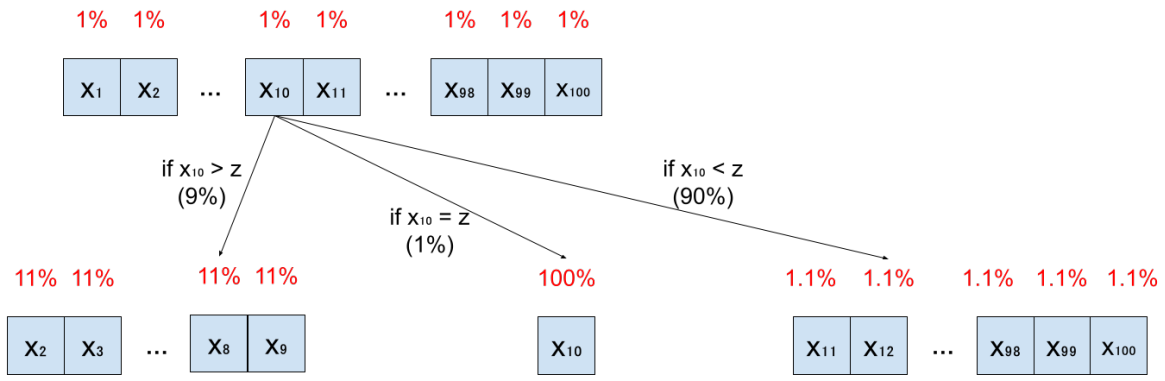


Figure 6.8 Posterior distributions of the location of target z within the 100-element list upon inspecting the tenth element. In each scenario, the probability that z is at a given location is updated accordingly.

SIDEBAR **SIDEBAR**

Make sure that you attempt the entropy computations yourself to understand how we are getting these numbers.

Finally, we take the average of these entropies using the corresponding probabilities: *0.09 (3.17) + 0.01 (0) + 0.9 (6.49) = 6.13*. This number presents the expected posterior entropy, that is, the expected posterior uncertainty we have about L, the location of z, after inspecting the tenth element x_{10}.

Compared to the same number for the first element x_1, 6.56, we conclude that looking at x_{10} on average gives us more information about L than looking at x_1. In other words, inspecting x_{10} is the information-theoretically better decision.

But what is the information-theoretically *optimal* decision, the one that gives us the most information about L? To determine this, we simply repeat the same computation we just performed on x_1 and x_{10} for the other locations in the list and pick out the one giving the lowest expected posterior entropy.

Figure 6.9 shows this quantity, the expected posterior entropy in the location of our target, as a function of the location that we choose to inspect. The first thing we notice is the symmetry of the curve: looking at the last location gives us the same expected posterior entropy (uncertainty) as looking at the first; similarly, the tenth and the ninetieth locations give the same amount of information, and so on.

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